

9.1. The Basics

Formulas and functions are what make Excel more than just a printable grid of columns and rows. As explained in Chapter 3, you create a formula by using mathematical and calculation operators in combination with numbers or cell references ($=5+3$, $=A1+B6$, etc.) or by using pre-built functions that make it easier to do complex calculations in a single step ($=SUM(A2:A10)$, etc.).

All formulas and functions start with the “=” sign. This indicates to Excel that what follows is a calculation. Calculations are done using operators. In many cases the operators themselves are sufficient to obtain the required results, without having to depend on functions. There are four types of operators in Excel 2007: arithmetic, comparison, text, and reference.

Depending on the operator, the order of calculation may also vary. Each operator has a different level of precedence: in a formula where more than one operator is used, Excel will calculate the result in a pre-determined sequence.

For example, the formula “ $=2+5*3/-4$ ” will first calculate 3 divided by -4, since negative values have the highest precedence. This will be followed by the multiplication and then the addition. Hence the result will be 1.75 and not -5.25, which would have been the result if you had used a simple left-to-right calculation order.



The operator types

Order	Symbol
1	- (negation)
2	%
3	^
4	* and /
5	+ and -
6	&
7	=, <, >, <=, >=, <>

Operator calculation order